

DA6360 WS 3

1. linear approximant $\gamma: \mathbb{R} \rightarrow \mathbb{R}$ around $x_0 = 0$
 $|_{x_0=0}$

$$\begin{aligned} L(x) &= f(0) + f'(0)(x-0) \\ &= f(0) + f'(0)x \\ f(0), f'(0) \end{aligned}$$

(a) $\sin x$ $\sin 0 = 0$ $\cos 0 = 1$

$$L(x) = 0 + 1(x) = x$$

(b) $f(x) = 3x^4 + 4x^3 + 8$

$$f(0) = 8, \quad f'(0) = 12x^3 + 12x^2 + 0 = 0$$

$$L(x) = 8 + 0(x) = 8$$

(c) $f(x) = e^x + 2x$

$$f(0) = e^0 + 2(0) = 1$$

$$f'(x) = e^x + 2 \Rightarrow f'(0) = 1 + 2 = 3$$

$$L(x) = 1 + 3x = 3x + 1$$

(d) $f(x) = \ln(2+x), \quad x \in (-2, \infty)$

$$f(0) = \ln(2)$$

$$f'(0) = \frac{1}{2+x} = \frac{1}{2}$$

$$L(x) = \ln 2 + \frac{x}{2}$$

(e) $f(x) = 2x^2, \quad f(0) = 0, \quad f'(0) = 0$

$$L(x) = 0$$

(f) $f(x) = \sin(x^2)$

$$f(0) = 0$$

$$f'(x) = \cos(x^2) \cdot 2x$$

$$f'(0) = 0$$

$$L(x) = 0$$

$$L(x) = f(1) + f'(1)(x-1)$$

(i) linear approximation $x_0 = 1$

$$\begin{array}{ll} \text{(a)} & \sin(x) \quad \sin(1) \\ & \cos(x) \quad \cos(1) \end{array}$$

$$L(x) = \sin 1 + \cos 1(x-1)$$

$$\begin{array}{ll} \text{(b)} & 2x^4 + 4x^3 + 8 \quad f(1) = 15 \\ & 12x^3 + 12x^2 + 8 \quad f'(1) = 24 \end{array}$$

$$L(x) = 15 + 24(x-1)$$

$$\begin{array}{ll} \text{(c)} & e^x + 2x \quad f(1) = e + 2 \\ & e^x + 2 \quad f'(1) = e + 2 \end{array}$$

$$L(x) = (e+2) + (e+2)(x-1)$$

$$\begin{array}{ll} \text{(d)} & \ln(2+x) \quad f(1) = \ln 3 \\ & \frac{1}{2+x} \quad f'(1) = \frac{1}{3} \end{array}$$

$$L(x) = \ln 3 + \frac{(x-1)}{3}$$

$$\begin{array}{ll} \text{(e)} & 2x^2 \quad f(1) = 2 \\ & 4x \quad f'(1) = 4 \end{array}$$

$$L(x) = 2 + 4(x-1)$$

$$= 2 + 4x - 4 = 4x - 2$$

$$\begin{array}{ll} \text{(f)} & \sin x^2 \quad f(1) = \sin 1 \\ & \cos x^2 \cdot 2x \quad f'(1) = 2 \cos 1 \end{array}$$

$$L(x) = \sin 1 + 2 \cos 1(x-1)$$

(ii) Quadratic approx. at $x_0 = 0$

$$T_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2$$

(a) $\sin(x)$ $\sin(0) = 0$

$\cos(x)$ $\cos(0) = 1$

$-\sin(x)$ $-\sin(0) = 0$

$$T_2(x) = 0 + x + 0 = x$$

(b) $3x^4 + 4x^3 + 8$

$f(0) = 8$

$12x^3 + 12x^2$

$f'(0) = 0$

$36x^2 + 24x$

$f''(0) = 0$

$$T_2(x) = 8 + 0 + 0 = 8$$

(c) $e^x + 2x$

$f(0) = 1$

$e^x + 2$

$f'(0) = 1 + 2 = 3$

e^x

$f''(0) = 1$

$$T_2(x) = 1 + 3x + \frac{1x^2}{2} = 1 + 3x + \frac{x^2}{2}$$

(d) $\ln(2+x)$

$\ln 2$

$\frac{1}{2+x} = (2+x)^{-1}$

$\frac{1}{2}$

$-1(2+x)^{-2}$

$= -\frac{1}{(2+x)^2}$

$= -\frac{1}{4}$

$$T_2(x) = \ln 2 + \frac{x}{2} + \left(-\frac{x^2}{8} \right)$$

$$(e) \quad 2x^2$$

$$4x$$

$$4$$

$$f(0) = 0$$

$$f'(0) = 0$$

$$f''(0) = 4$$

$$T_2(x) = 0 + 0 + \frac{4x^2}{2} = 2x^2$$

$$(f) \quad \sin(x^2)$$

$$\cos(x^2) \cdot 2x$$

$$2\cos x^2 + (-\sin x^2 \cdot 2x \cdot 2x)$$

$$\hookrightarrow 2\cos x^2 - \sin x^2 \cdot 4x^2$$

$$f(0) = 0$$

$$f'(0) = 0$$

$$f''(0) = 2$$

$$T_2(x) = 0 + 0 + \frac{2x^2}{2} = x^2$$

(iii) Quadratic approx. at $x_0 = 1$

$$T_2(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2}(x-1)^2$$

(a)	$\sin x$	$\sin 1$
	$\cos x$	$\cos 1$
	$-\sin x$	$-\sin 1$

$$T_2(x) = \sin 1 + \cos 1(x-1) - \frac{\sin 1(x-1)^2}{2}$$

(b)	$3x^4 + 4x^3 + 8$	$f(1) = 15$
	$12x^3 + 12x^2$	$f'(1) = 24$
	$36x^2 + 24x$	$f''(1) = 60$

$$T_2(x) = 15 + 24(x-1) + \frac{60}{2}(x-1)^2$$

$$= 15 + 24x - 24 + 30(x-1)^2$$

(c)	$e^x + 2x$	$f(1) = e + 2$
	$e^x + 2$	$f'(1) = e + 2$
	e^x	$f''(1) = e$

$$T_2(x) = (e+2) + (e+2)(x-1) + \frac{e}{2}(x-1)^2$$

(d)	$\ln(2+x)$	$f(1) = \ln 3$
	$\frac{1}{2+x}$	$f'(1) = \frac{1}{3}$
	$-\frac{1}{(2+x)^2}$	$f''(1) = -\frac{1}{9}$

$$T_2(x) = \ln 3 + \frac{x}{3} + \left(-\frac{(x-1)^2}{18} \right)$$

$$(e) \quad 2x^2$$

$$4x$$

$$4$$

$$f(1) = 2$$

$$f'(1) = 4$$

$$f''(1) = 4$$

$$T_2(x) = 2 + 4x + \frac{4x^2}{2}$$

$$= 2x^2 + 4x + 2$$

$$(f) \quad \sin x^2$$

$$2x \cdot \cos(x^2)$$

$$2 \cos x^2 + (2x \cdot 2x \cdot -\sin x^2)$$

$$2 \cos x^2 + (-4x^2 \sin x^2)$$

$$f(1) = \sin 1$$

$$f'(1) = 2 \cos 1$$

$$f''(1) = 2 \cos 1 + (-4 \sin 1)$$

$$T_2(x) = \sin 1 + 2 \cos 1 (x-1) + \frac{(2 \cos 1 - 4 \sin 1)(x-1)^2}{2}$$

$$2. \quad f(x) = \ln(2+x), \quad x \in [-2, \infty)$$

$$x_0 = 0$$

$$T_4(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$$

$$f(x) = \ln(2+x)$$

$$f'(x) = \frac{1}{2+x} = (2+x)^{-1}$$

$$f(0) = \ln 2$$

$$f'(0) = \frac{1}{2}$$

$$f''(x) = -\frac{1}{(2+x)^2} = -(2+x)^{-2}$$

$$f''(0) = -\frac{1}{4}$$

$$f'''(x) = 2 \frac{1}{(2+x)^3} = 2(2+x)^{-3}$$

$$f'''(0) = \frac{1}{4}$$

$$f^{(4)}(x) = -6 \frac{1}{(2+x)^4} = -6(2+x)^{-4}$$

$$f^{(4)}(0) = -\frac{6}{16}$$

$$T_4(x) = \ln 2 + \frac{1}{2} + \left(\frac{-1}{4 \times 2} x^2 \right) + \frac{x^3}{24} - \frac{6x^4}{24 \times 16}$$

$$= \ln 2 + \frac{1}{2} - \frac{x^2}{8} + \frac{x^3}{24} - \frac{6x^4}{384}$$

$$3. \quad f(x) = \ln(2+x)$$

$$f'(x) = \frac{1}{2+x}$$

$$g(x) = a + bx + ce^x$$

$$\ln 2$$

$$x_0 = 0$$

$$f''(x) = -\frac{1}{(2+x)^2}$$

$$\frac{1}{2}$$

$$-\frac{1}{4}$$

$$g(x) = a + bx + ce^x$$

$$a + c$$

$$g'(x) = b + ce^x$$

$$b + c$$

$$g''(x) = ce^x$$

$$c$$

$$C = -\frac{1}{4}$$

$$b + C = \frac{1}{2}$$

$$b - \frac{1}{4} = \frac{1}{2} \Rightarrow b = \frac{1}{2} + \frac{1}{4} = \frac{2+1}{4} = \frac{3}{4}$$

$$a + C = \ln 2$$

$$a - \frac{1}{4} = \ln 2 \Rightarrow a = \ln 2 - \frac{1}{4}$$

4. $f(x) = x^4 - x^2$

(a) stationary pts. are those where $f'(x) = 0$

$$f'(x) = 4x^3 - 2x = 0$$

$$4x^3 - 2x = 2x(2x^2 - 1) = 0$$

$$x = 0$$

$$2x^2 - 1 = 0$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = 0 \Rightarrow f(0) = 0$$

$$x = \pm \frac{1}{\sqrt{2}} \Rightarrow f\left(\pm \frac{1}{\sqrt{2}}\right) = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

$$(0, 0), \left(\frac{1}{\sqrt{2}}, -\frac{1}{4}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{4}\right)$$

(b) local / global minima / maxima

$$f''(x) = 12x^2 - 2$$

$$x=0 \quad f''(0) = -2 < 0$$

↳ local maxima

$$f(0) = 0$$

↳ global maxima

$$x = \pm \frac{1}{\sqrt{2}} \quad f''\left(\pm \frac{1}{\sqrt{2}}\right) = 4 > 0$$

both pts. are local minima,
also they are global minima.

(c) first order, $x=1$

$$f(x) = x^4 - x^2$$

$$f(1) = 1 - 1 = 0$$

$$f'(x) = 4x^3 - 2x$$

$$f'(1) = 4 - 2 = 2$$

$$\begin{aligned} T_1(x) &= f(1) + f'(1)(x-1) \\ &= 2(x-1) \\ &= 2x - 2 \end{aligned}$$

(d) second order, $x=-1$

$$f''(x) = 12x^2 - 2$$

$$f(-1) = 1 - 1 = 0$$

$$f'(-1) = -4 + 2 = -2$$

$$f''(-1) = 12 - 2 = 10$$

$$\begin{aligned} T_2(x) &= f(-1) + f'(-1)(x+1) + \frac{f''(-1)}{2}(x+1)^2 \\ &= 0 + (-2)(x+1) + \frac{10}{2}(x+1)^2 \\ &= -2(x+1) + 5(x+1)^2 \end{aligned}$$

5. $f(x) = x^4 - x^2$

as posed in the prev. ques.

6. $x_{k+1} = x_k - \eta f'(x_k)$

quadratic $0 < \eta < \frac{1}{a}$

$$\rightarrow a(x-b)^2$$

$$a(x^2 + b^2 - 2bx)$$

$$ax^2 + ab^2 - 2abx$$

(a) $f(x) = 5x^2$
 $f'(x) = 10x$

$$x_{k+1} = x_k - \eta f'(x_k)$$

$$x_{k+1} = x_k - \eta (10x_k)$$

$$x_{k+1} = x_k (1 - \eta 10)$$

$$(1 - 10\eta) < 1$$

$$0 < \eta < \frac{1}{5}$$

$$(b) \quad f(x) = \frac{x^2}{100}$$

$$0 < \eta < 100$$

$$f'(x) = \frac{2x}{100} = \frac{x}{50}$$

$$x_{k+1} = x_k - \eta \left(\frac{x_k}{50} \right)$$

$$(c) \quad f(x) = (x-5)^2$$

$$= x^2 + 25 - 10x$$

$$0 < \eta < 1$$

$$f'(x) = 2x - 10$$

$$x_{k+1} = x_k - \eta (2x_k - 10)$$

if constant < 1 , you converge
 $c = 1$, do not settle
 $c > 1$, diverge.